Homework #7

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November 1, 2026

Exercise 1

Large-step semantics under call-by-name:

$$\frac{e_1 \Downarrow \lambda x.\, e \quad e\{e_2/x\} \Downarrow v}{e_1\, e_2 \Downarrow v} \qquad \frac{}{\lambda x.\, e \Downarrow \lambda x.\, e}$$

Exercise 2

(a)

$$\mathit{is_leaf} \;:=\; \lambda t.\, t\ (\lambda _ _ _.\, \mathsf{FALSE})\ \mathsf{TRUE}$$

(b)

$$\mathit{root} \;:=\; \lambda t.\, t\ (\lambda _\, v\, _.\, v)\ 0$$

(c)

$$\mathit{nodes} \;:=\; \lambda t.\, t\ (\lambda s_1\, _\, s_2.\, \mathsf{PLUS}\ s_1\ (\mathsf{PLUS}\ s_2\ 1))\ 0$$

(d)

$$\mathit{height} \;:=\; \lambda t.\, t\ (\lambda h_1\, _\, h_2.\, \mathsf{PLUS}\ 1\ (\mathsf{MAX}\ h_1\ h_2))\ 0$$

Exercise 3

(a)

$$\begin{aligned}
\mathcal{ST}[\![()]\!]\, s &= ((),\, s) \\
\mathcal{ST}[\![\mathsf{get}]\!]\, s &= (s,\, s) \\
\mathcal{ST}[\![\mathsf{set}(e)]\!]\, s &= \big((),\, \#1\,(\mathcal{ST}[\![e]\!]\, s)\big) \\
\mathcal{ST}[\![\lambda x.\, e]\!]\, s &= \big(\lambda x.\, \lambda s_1.\, \mathcal{ST}[\![e]\!]\, s_1,\ s\big) \\
\mathcal{ST}[\![e_1\, e_2]\!]\, s &= \\
&\qquad \mathsf{let}\ ste_1 = \mathcal{ST}[\![e_1]\!]\, s\ \mathsf{in} \\
&\qquad \mathsf{let}\ ste_2 = \mathcal{ST}[\![e_2]\!]\, (\#2\, ste_1)\ \mathsf{in} \\
&\qquad (\#1 ste_1)\, (\#1 ste_2)\, (\#2 ste_2)
\end{aligned}$$

(b)

$$\begin{aligned}\mathcal{ST}[\![\mathsf{while}\ e_1\ \mathsf{do}\ e_2]\!]\,s\ =\ \Big((),\\ Z\,\big(\lambda f.\,\lambda s.\\ \quad \mathsf{let}\ ste_1 = \mathcal{ST}[\![e_1]\!]\,s\ \mathsf{in}\\ \quad \mathsf{if}\ (\#1\ ste_1)\ \mathsf{then}\ \big(f\ (\#2\,(\mathcal{ST}[\![e_2]\!]\,(\#2\ ste_1)))\big)\\ \quad \mathsf{else}\ (\#2\ ste_1)\big)\ s\Big)\end{aligned}$$

where Z is the call-by-value fixed-point combinator

$$Z \triangleq \lambda f.\,(\lambda x.\,f\,(\lambda y.\,x\,x\,y))\,(\lambda x.\,f\,(\lambda y.\,x\,x\,y)).$$

AI Disclosure

1. Claude LLM
2. Asked to convert handwriting to LaTeX
3. Saved time writing

Debriefing

1. 3 to 4 hours
2. moderate-difficult
3. yes
4. 85%